

EP 301: Engineering Mathematics-2

Course Code: EP 301

Course Name: Engineering Mathematics-2

L-T-P-C: 3-1-0-4

Prerequisite: IC 110, IC 111 and/or faculty consent; Out of PH511 and EP301, a student is allowed to take only one course.

Students intended for: B.Tech.

Elective or Core: Core

Approval: 28th Senate

Course content

- **Module I**

Linear vector spaces, Gram-Schmidt orthogonalization, Self-adjoint, Unitary, Hermitian, Non-Hermitian Operators, transformation of operators, eigenvalue equation, Hermitian matrix diagonalization; with focus on application to physical problems. [8 Lectures]

- **Module II**

Second order Linear ODE, Series Solution- Frobenius Method, Inhomogeneous linear ODE. Sturm Liouville equation Hermitian operators-eigenvalue problem; with focus on application to physical problems. [7 Lectures]

- **Special functions**

Bessel, Neumann, Henkel, Hermite, Legendre, Spherical Harmonics, Laguerre, Gamma, Beta, Delta functions, with focus on application of these functions to physical problems. [10 Lectures]

- **Module IV**

Complex analysis, Cauchy- Riemann conditions, Cauchy's Integral theorem, Laurent expansion, Singularities, Calculus of residues, evaluation of definite integrals, Method of steepest descent, saddle point. [11 Lectures]

- **Module V**

Partial differential equations and introduction to Green's functions. [6 Lectures]

Text Books

1. Arfken and Weber, **Mathematical methods for physicists**, 6th Edition, Elsevier Academic Press, 2015.
2. Mary L Boas, **Mathematical Methods in Physical Sciences**, 3rd Edition, Wiley 2011

Reference Books:

1. K. F. Riley and M. P. Hobson, **Mathematical Methods for Physics and Engineering: A Comprehensive Guide**, Cambridge India South Asian Edition, 2009.
2. J. Mathews and R. L. Walker, **Mathematical Methods for Physicists**, New edition, Imprint, 1973.



3. F. W. Byron and R. W. Fuller, **Mathematics of Classical and Quantum Physics**, New edition, Dover Publication, 1992.
4. P. M. Morse and H. Freshbach, **Methods of theoretical Physics Vol. I and II**, McGraw-Hill, 1953.
5. E. Kreyszing, **Advanced Engineering Mathematics**, 10th edition, Wiley India Private Limited, 2003.
6. Philippe Dennery and Andre Krzywicki, **Mathematics for Physicists**, Dover Publications Inc. 1996.